

device selection)
top-M device routing

communication ~~loss~~ ^{balancing loss} - balancing both communication in and out

V3 (G1B-37 active params)

shared CI) + fine-grained (258) experts, 8 active

new things:

Sigmoid + Softmax topK + topM

Aux-loss-free + seq-wise aux

Question: what else do you need to make 175K MoE V3?

MLA: Multihead, Latent Attention.

#1 Basic idea: express the Q, K, V, as a functions of a lower-dim, 'latent' attention.

#2 Benefits: when KV-caching, we only need to store C_e^{KV} , which can be much smaller, W^{OK} can be merged into the Q projection

#3 complexity: rope conflicts with MLA-style caching

#4 MTP: have small, lightweight models that predict multiple steps ahead

⇒ target LLM / draft LLM

MOE Summary

- #1 MOEs take advantage of sparsity - not all inputs need the full model
 - #2: Discrete routing is hard, but top-K heuristics seem to work
 - #3: Lots of empirical evidence now that MOEs work, and cost-effective
-